

Nexus Dynamics Group

Business Intelligence Case Study

An end-to-end analytics solution for financial performance, workforce efficiency, supplier risk, inventory exposure and regional sales.

ROLE	Power BI Developer & Data Analyst
TOOLS	Power BI • Power Query • DAX • Data Modelling
SCOPE	5 source datasets • 41,740 records • 4 report pages
DELIVERABLE	Interactive executive dashboard and management recommendations

Wandile Bingo

September 2026

Executive Summary

From fragmented operational data to one decision-ready management view.

This project converts five supplied CSV datasets into a structured Power BI solution covering finance, people, operations, suppliers and sales. The work included data preparation, relationship design, DAX measures, interactive report development and a management-level interpretation of the results.

Business objective

Give management a single place to monitor profitability and liquidity, identify workforce and supplier risks, control critical stock, and compare sales and returns by region.

Metric	Result	Decision relevance
Gross profit margin	50.04%	Overall profitability baseline
Current ratio	1.14	Adequate but narrow liquidity cover
Supplier score	74.31%	Portfolio-wide weighted performance
Critical stock	771 records	Immediate replenishment exposure
Top 10 customer sales	R13.01m	High-value customer concentration
Rolling 3-month net sales	R32.55m	Recent revenue momentum

Project scope

- Prepared and validated 41,740 records: Financials (240), HR (10,000), Inventory (5,000), Sales (25,000) and Supplier (1,500).
- Designed shared Date, Product and Supplier dimensions to support consistent filtering across related fact tables.
- Built four report pages for KPIs, financial performance, inventory and supplier analysis, and sales and workforce analysis.

Data Preparation & Model Architecture

A simple star-style model was used to make filters predictable and measures reusable.

- Standardised date fields and removed time components so relationship keys matched across tables.
- Applied suitable numeric, percentage, currency and date data types in Power Query and the semantic model.
- Created unique Product, Supplier and Date dimensions, then connected them to the relevant transaction tables using one-to-many, single-direction relationships.
- Validated the model by confirming that year, product and department selections changed the intended visuals without creating ambiguous filter paths.

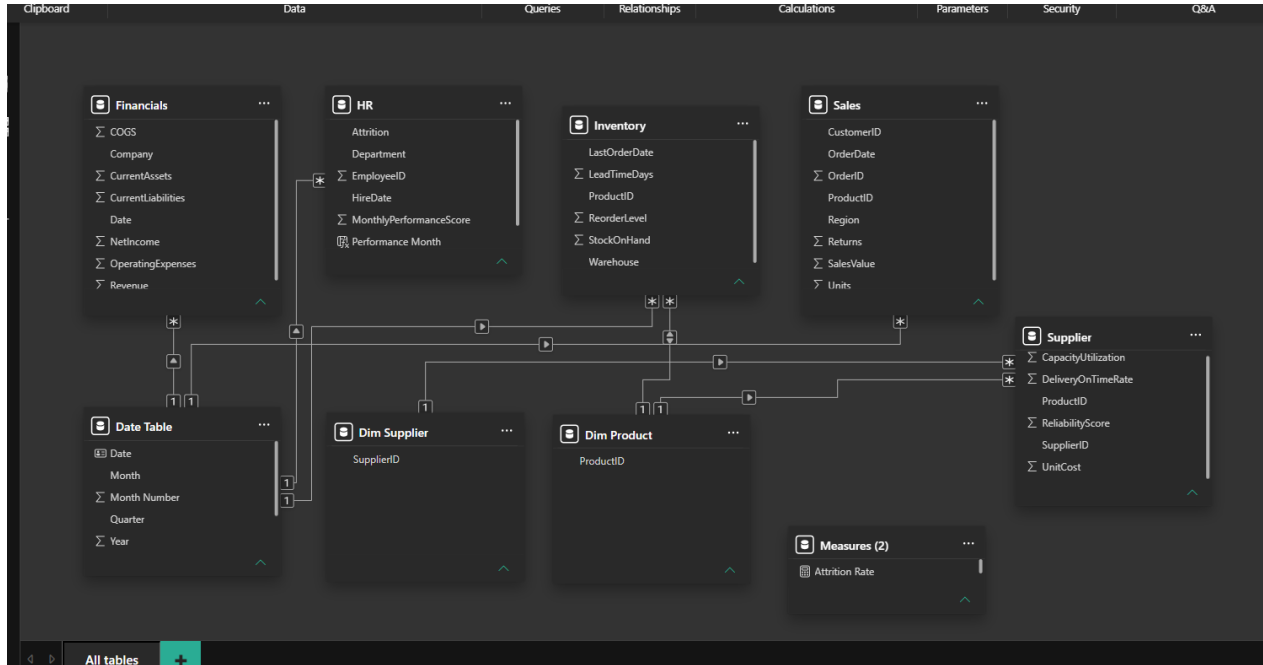


Figure 1. Semantic model used to support cross-table analysis and consistent filtering.

Modelling lesson

The initial flat charts occurred because category fields and values came from tables without an effective filter path. Correct keys, matching data types and active one-to-many relationships resolved the repeated values and restored slicer behaviour.

KPI & Measure Framework

Reusable DAX measures translate the model into management indicators.

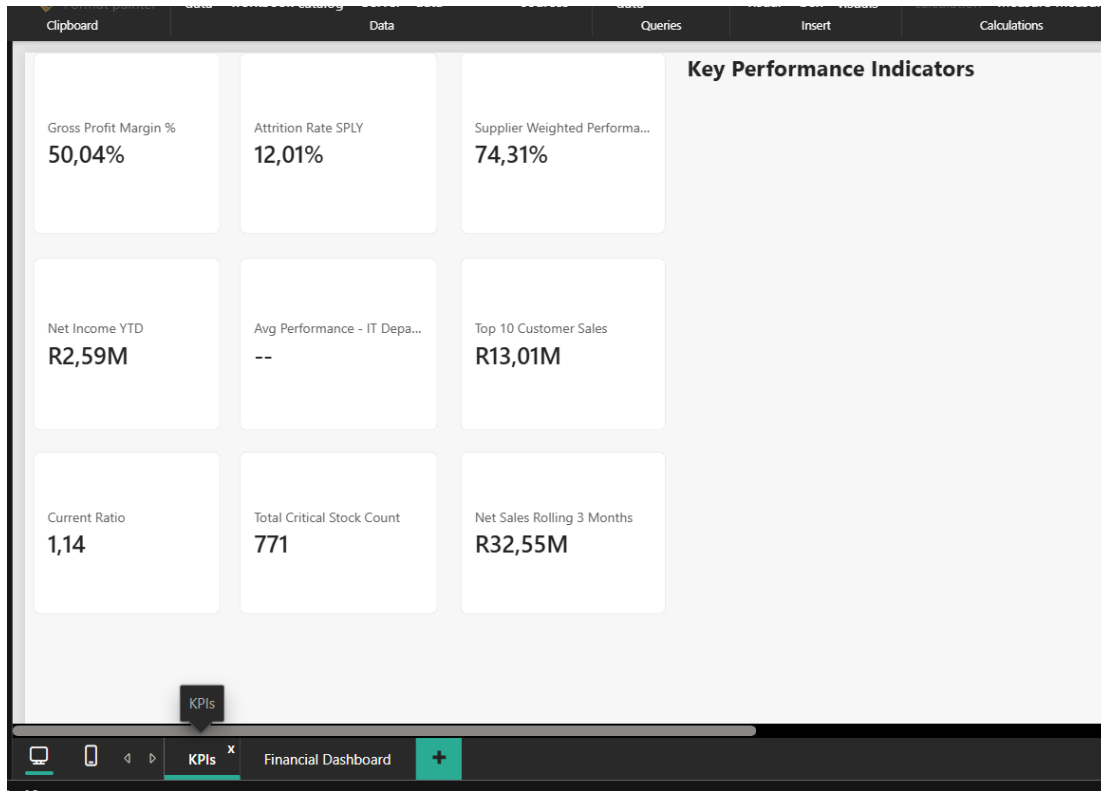


Figure 2. Executive KPI page consolidating financial, people, supplier, inventory and sales indicators.

Metric	Result	Decision relevance
Financial	Margin, YTD income, current ratio	Profitability and liquidity
People	Attrition SPLY, average performance	Retention and workforce effectiveness
Operations	Critical stock count	Items below reorder level
Supplier	Weighted performance score	Reliability, delivery and capacity
Sales	Top 10 sales, rolling 3 months	Customer value and recent momentum

Data-quality exception handled transparently

The IT performance card is blank because no IT department exists in the HR source. In a production version, the card would use the selected department or display an explicit “No data available” state rather than implying a calculation failure.

Financial Performance Dashboard

Monthly performance, company comparison and year-based filtering in one view.

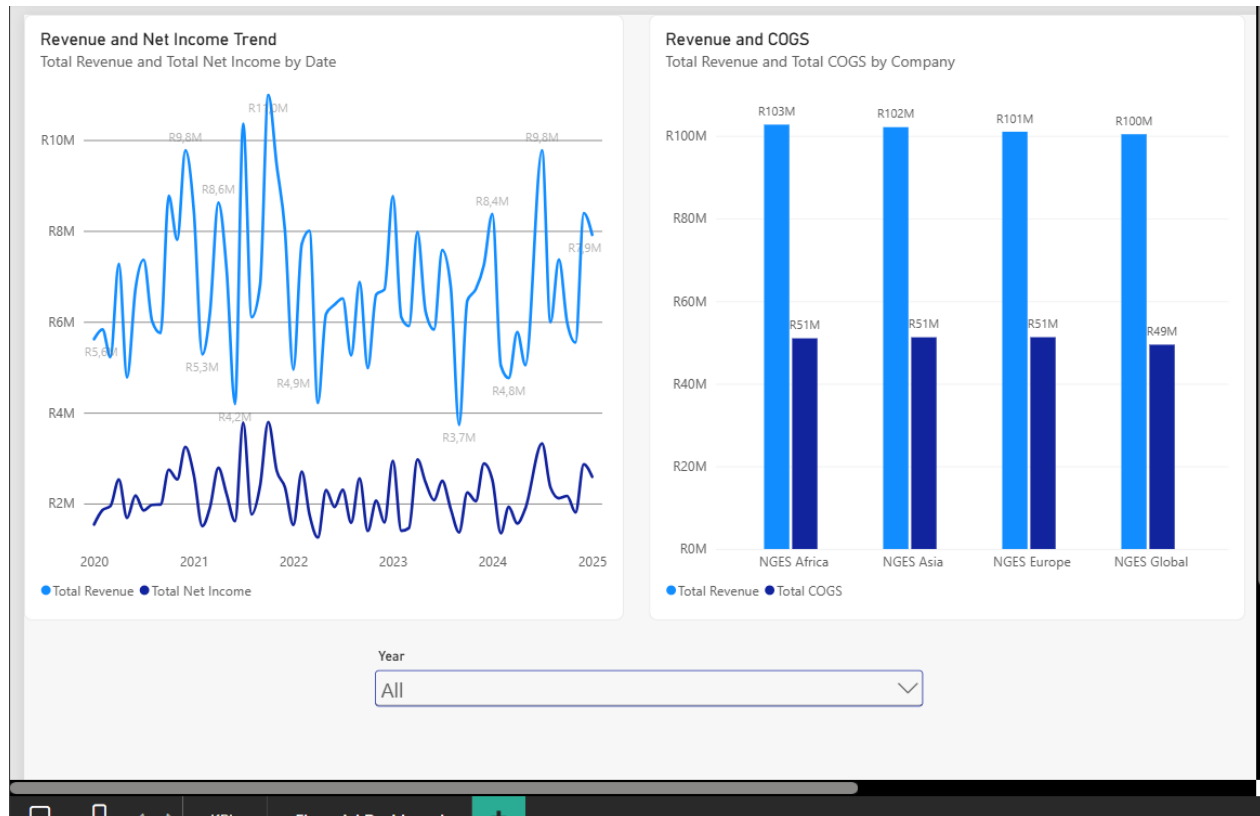


Figure 3. Financial dashboard: revenue and net-income trend, revenue versus COGS, and year selection.

What the dashboard shows

- Net income increased from R22.95m in 2022 to R26.24m in 2023, then eased to R23.88m in 2024.
- January 2025 contributed R2.59m; it is a partial-year value and should not be compared directly with full-year totals.
- Revenue remains close to R100m–R103m per company, while COGS is approximately R49m–R51m, consistent with the 50.04% overall gross margin.

Management use

The year slicer and cross-filtering help management isolate periods and companies that drove changes in revenue, costs and net income, while avoiding misleading comparisons between full and partial years.

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Inventory & Supplier Risk Dashboard

Supplier quality and replenishment risk are viewed together to support operational decisions.

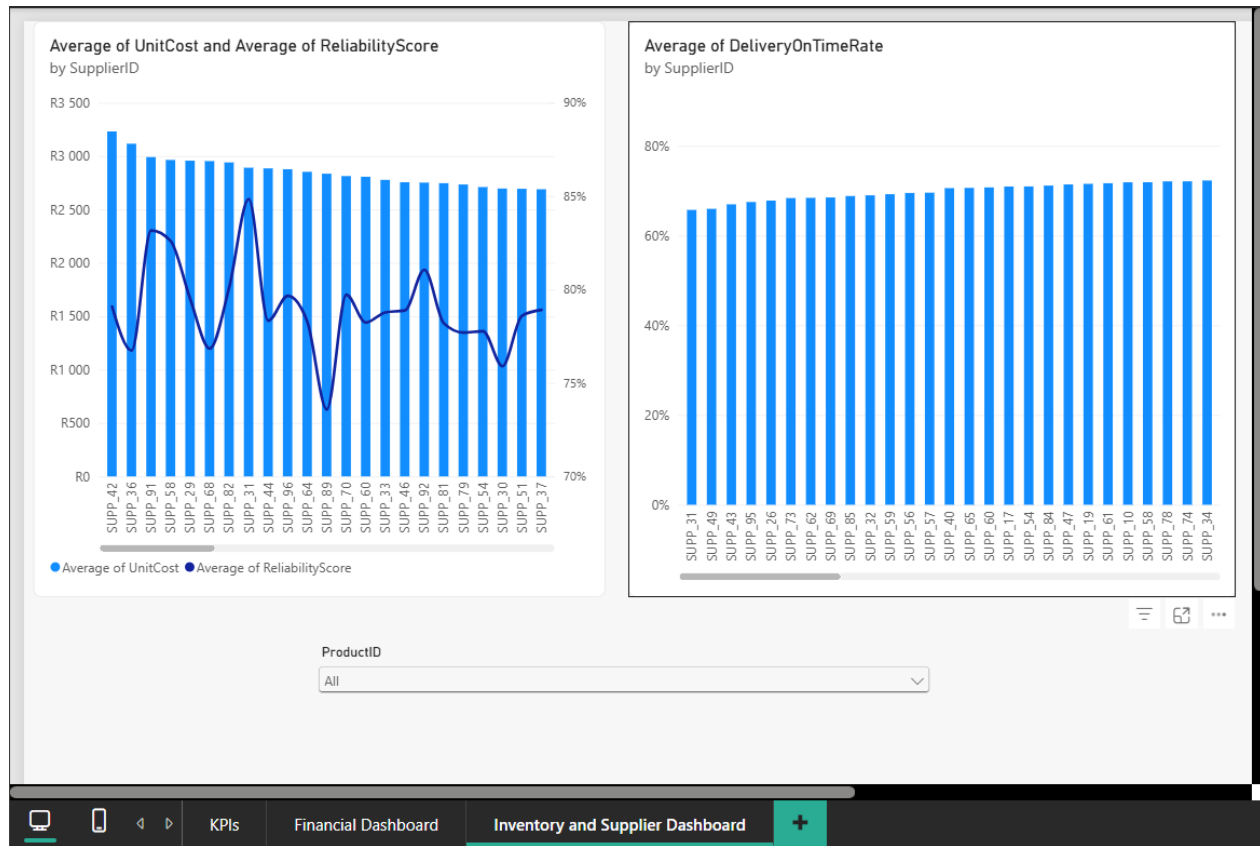


Figure 4. Supplier cost, reliability and delivery performance with product-level filtering.

What the dashboard shows

- The overall weighted supplier score is 74.31%, combining reliability (40%), delivery (35%) and capacity utilisation (25%).
- SUPP_78 is the highest-risk supplier at 69.08%; its 57.89% capacity score is the largest weakness.
- There are 771 critical-stock records overall, including 191 in the North Warehouse where stock on hand is below the reorder level.

Interpretation note

The lowest bar on the delivery-only chart does not have to be the lowest weighted supplier. The weighted measure evaluates three risk dimensions, so it provides the broader supplier-risk ranking.

Sales & Workforce Dashboard

Regional sales and returns sit alongside workforce performance for a broader operating view.

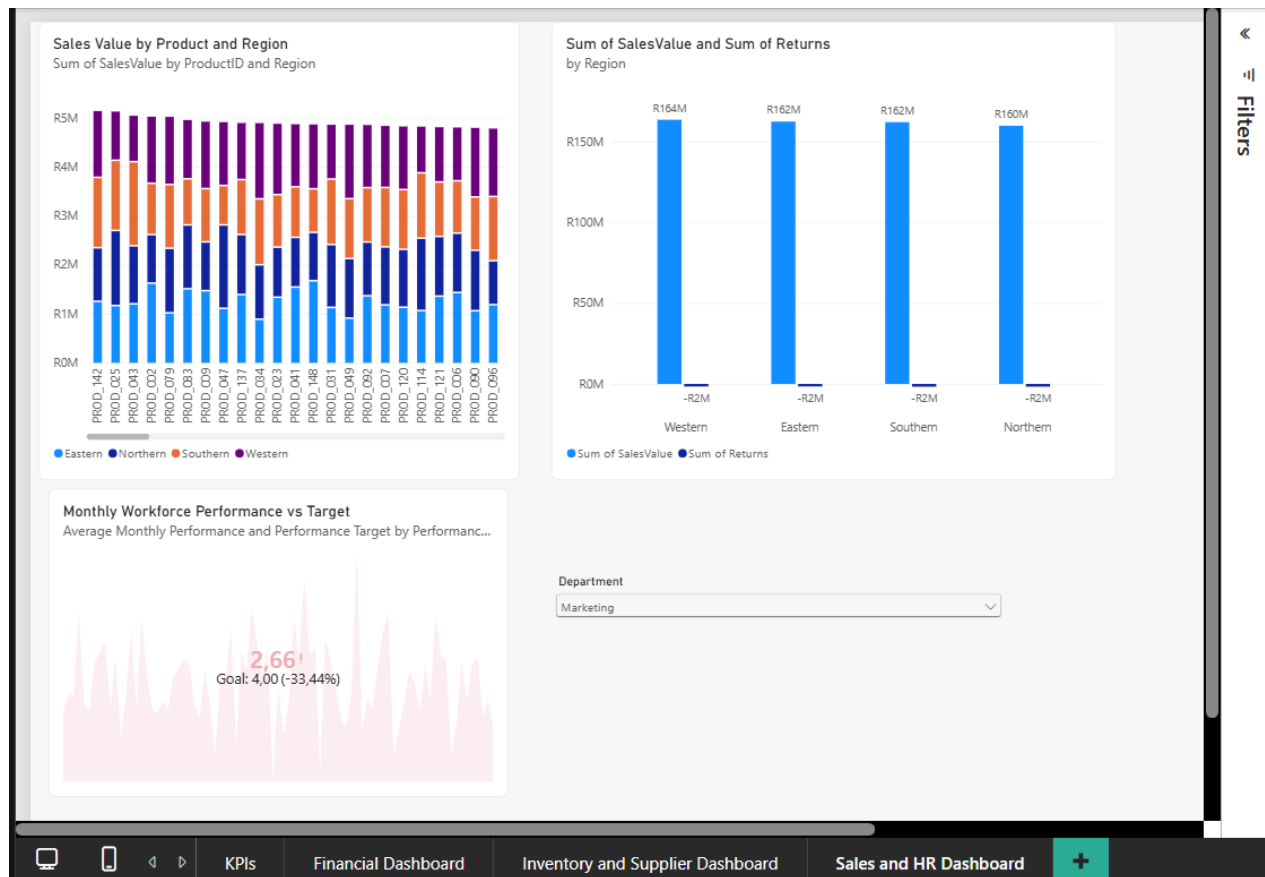


Figure 5. Sales mix, regional return impact and department-level workforce performance.

What the dashboard shows

- Regional sales are balanced: Western R163.55m, Eastern R162.42m, Southern R162.07m and Northern R159.85m.
- Northern has the highest negative-return ratio at approximately 1.03%, narrowly above Southern at 1.026%.
- Marketing records the highest attrition rate at 13.28% and a performance score of 2.66 against the 4.0 target when selected.

Interaction design

The Department slicer is intentionally scoped to HR analysis. It should change the workforce KPI, but not unrelated sales visuals because the Sales and HR tables do not share a business key beyond separate analytical contexts.

Management Findings

The dashboards highlight four linked areas requiring management attention.

1. Profitability is positive, but momentum is uneven

The business remains profitable, with a 50.04% gross margin. Net income improved by about 14.3% from 2022 to 2023, but declined by about 9.0% in 2024. The current ratio improved from 1.14 in 2022 to 1.20 in 2023 before easing to 1.17 in 2024. This indicates adequate short-term cover, although the liquidity buffer weakened slightly.

2. Retention pressure is concentrated

Marketing (13.28%), Logistics (12.82%) and Admin (11.96%) have the highest attrition rates. Attrition increased from 11.04% for the 2023 hire cohort to 11.82% for 2024, while average performance declined from 3.00 to 2.95. The pattern supports focused departmental action rather than a broad, undifferentiated HR response.

3. Supplier capacity and critical stock create service risk

SUPP_78 has the lowest weighted score at 69.08%, driven mainly by low capacity utilisation performance. Combined with 191 critical-stock records in the North Warehouse, this creates exposure to stock-outs, delayed fulfilment and avoidable revenue loss.

4. Sales are balanced, but customer value and returns need targeting

The Top 10 customers generate R13.01m and all transact across all four regions. By primary spending region, four are Southern, four Northern, one Eastern and one Western. Management should protect the stronger Northern and Southern relationships while investigating the Northern return ratio before increasing promotional spend.

Overall conclusion

Nexus Dynamics is profitable and liquid, but stable performance depends on tighter supplier-capacity controls, faster action on critical inventory, focused retention measures and better analysis of Northern returns.

Recommendations & Portfolio Summary

Two practical actions address the most immediate operational and people risks.

Recommendation 1 — Stabilise supply and inventory risk

Place SUPP_78 on a 90-day improvement plan with agreed capacity, reliability and delivery targets. Identify critical products linked to the supplier, approve a backup source where practical, and review reorder levels and safety stock for the 191 North Warehouse exceptions. Track progress through a weekly exception report until critical stock and supplier-risk indicators improve.

Recommendation 2 — Target retention and performance

Begin with Marketing, Logistics and Admin. Use stay interviews, exit analysis, workload and pay reviews, and manager-level action plans during the first 60 days. Track progress against the 2023 attrition baseline of 11.04% and the 4.0 performance target, with named owners and quarterly reviews.

Expected business value

- Lower probability of stock-outs and service disruption.
- More dependable supplier performance and clearer escalation paths.
- Lower replacement costs and more consistent workforce performance.
- More disciplined regional marketing and returns management.

Capabilities demonstrated

DATA PREPARATION	DATA MODELLING	DAX	REPORT DESIGN
Power Query cleaning and type validation	Dimension tables and filter relationships	Time intelligence, ranking and weighted measures	Interactive dashboards and management narrative

Portfolio positioning

This case study is presented as an independent, simulated business-intelligence project. It demonstrates the approach, technical decisions and management reasoning without representing the work as a paid client engagement.